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Secure data hiding in binary images using run-length pairs

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Abstract. The popularity of digital media is on the rise today. Besides multilevel images, video, and audio, binary document images are also digitized in many applications. These include legal documents, digital books, maps, pictures, and architectural drawings. Security and confidentiality of information transmitted through digital media are the primary concerns. This use of cryptography reveals a transfer of information as well as its existence and makes it easy for malpractices to occur. So information hiding is more promising as it focuses on imperceptibility. Hiding information in binary media is more difficult than color and grey media, as it causes more visual distortion due to limited two shades. Apart from this hiding technique needs to take care of hiding capacity and security of information hidden. An approach to hiding information in binary images which tries to maintain these concerns is presented in this paper. The message to be hidden is initially encrypted by key known to both, transmitter and receiver. This encrypted message is then represented as two dimensional sparse matrix and then transformed to one dimensional matrix where information of location of only black pixel '1' in sparse matrix is kept. The arithmetic compression algorithm is used to compress the sparse matrix. Black and white pixel run length (RL) pairs are utilized for

hiding information which satisfies the threshold set for RL pair length. On receiver side reverse process is applied to extract secret message. The encryption, sparse matrix representation and arithmetic data compression secures the information and enhances hiding capacity. Utilization of run length pair minimizes visual distortion. The results show that high capacity information is hidden with less distortion. As digital multimedia sources become more specialized, the information hidden within them is used for steganography and watermarking.

Keywords. Sparse matrix, encoding, binary image, information hiding, run-length

1.1 Introduction

As a result of the wide range of possible uses of digital documents, such as legal documents, certificates, digital books, and engineering drawings, there is great interest in their use. Further, sensitive documents such as faxes, insurance documents, and personal documents are digitized and stored. Securing the authenticity and integrity of digital documents is becoming increasingly relevant. Authenticating and detecting tempering and forgeries are therefore primary concerns. In order to mitigate these concerns, data hiding and watermarking have proven to be promising approaches. The use of internet is extensively done for digital data transfer over large distance that is use of internet is reducing efforts and takes negligible time for data transfer, but as use of Internet for data transfer increased chance of data being hacked. So security of data transfer from internet is major issue and focused research topic. Even though there are number of methods for secure data transfer like cryptography but they have their own pitfalls as it is self-declarative about existence of some important information. So message hiding in images does not disclose carrying of any information secretly. Passing secret message through color images is easier as they have 256 color ranges for each RGB color band as opposed to only two colours available in binary images. Its challenging to hide data in binary image maintaining visual distortion with high capacity. There are many techniques which uses spatial and frequency domain. One of the way is to use black and white pixel run length (RL) for hiding data.

1.2 Literature Survey

Data hiding, extraction and image recovery can be easily done using image encryption and support vector machine (SVM) [1]. For this purpose image blocks are used where in each block, the one-bit message can be embedded. Extraction can be carried out using support vector model (SVM) as it is trained. By adapting image block pattern, a small-block-based technique is proposed wherein information is hidden in [2]. This technique claims the highest data hiding rate. This technique applies the principle of Webb's law and also uses blind data extraction. The computational complexity in data hiding and extraction algorithm is very low. In [3], to check the effect of embedding and also to identify that the data is hidden a binary steganalysis scheme has been proposed. The distortion for 4 classes of patterns is calculated considering 1-shape pattern as the reference. Further, a 2-dimensional feature set consisting of only 2 patterns is computed. To reduce visual distortion, the combination theory has been applied. In order to increase hiding capacity and to reduce visual artifacts, a special matrix has been used. Each block is changed one pixel at a time and $\lceil \log_2 (m \times n + 1) \rceil$ bits are hidden into a block of size $m \times n$. It allows for a higher payload embedding under the same noise conditions [4]. In [5], block classification and local complexity calculations have been used. In this scheme each selected block is used to embed three bits data at most only two pixels are modified. In order to hide secret data, the complex region of the image is used so that it becomes difficult to detect the hidden data and also avoids grabbing attention from attackers. The texture property of binary images is considered in [6]. A similarity matching method based on texture is proposed as an efficient way to hide binary image data. In [7], new information security method is proposed. As compared to generally used LSB algorithm, the new proposed technique targets more unnoticeable and secure transmission of data that is secretly hidden. In [8], a novel scheme for authentication of binary image has been proposed, that has a small distortion in the cover image. In [9], in the data-embedding algorithm, hamming codes-based scheme has been used, where the ELSSM (Edge Line Segment Similarity Measure) algorithm is used for selecting and flipping pixels. A new multi-class steganalysis method was proposed in [10]. The capability of the proposed method is that it can identify the technique used for hiding data. The data hiding in binary text documents is proposed in [11] where the 8-connected boundary of character is used for embedding. By adding or deleting the center foreground pixel in a pair, a five-pixel boundary with a fixed set of pairs of patterns is utilized to embed data. After data hiding, an image quality

preserving scheme is presented in [12]. The imperceptibility after embedding is determined by comparing modified bits with their adjacent bits. The discrete cosine transform (DCT) domain is used in [13] to check for the possibility of embedding a watermark in binary images. Using the relationship within the block to identify "flippable" pixels in [14-16] results in minimal visual distortion. A method for detecting and hiding data in binary images to maintain visual quality is proposed in [17]. Several proposed schemes are also discussed that locate updated regions. An approach for preserving pixels' connectivity is presented in [18]. In a moving window, "flippability" is centered at pixel. [19] presents a two-layer blind binary image authentication scheme, in which the first layer enables the authentication, while the second layer identifies tampering locations. A scheme using morphological transforms is presented in [20]. The run-length (RL) histogram is monitored and modified to embed data using a reversible data hiding scheme. The influence of RL couples was used to conceal data from which black RL and white RL were formed. An examination and analysis of current watermarking and steganography technologies presented in [22], taken into account in the context of image processing in the spatial and transform domains. The regression analysis model used to accomplish the efficient de-fuzzification operation [23].

1.3 Gap Analysis

The fact that any pixel can be changed in a binary image makes hiding data in a binary image more challenging. Binary images have only two colors, so if a black pixel is changed to white or a white pixel to black, the image changes quite distinctly. The existing schemes, techniques, methods have limitations like limited data hiding capacity, visual distortion due to embedding, robustness, security of hidden data. These gaps can be filled as if the whole original image is utilized for data hiding, the capacity will be improved, if pattern of image in its original form is utilized as much as possible to hide data, visual distortion can be minimized, if data is encrypted before hiding, it will be secure as will not be readable or understandable.

1.4 Proposed technique

In proposed system the security of hidden data and the technique of data hiding is focused. 3 levels of security is provided by using encryption with key, representing sparse matrix in single column form and arithmetic data compression. Secret message is encrypted by a key which is known to both the sides. Then it is represented in two dimensional sparse matrix which is represented in one column form. The arithmetic compression algorithm is used to compress the secret message represented in above form. This encrypted data and compressed data is then embedded into the cover image using run-length pairs. In order to generate alternative black RL and white RL sequences, the binary image is scanned from left to right and from top to bottom. One RL pair is generated when a black RL is combined with its next white RL, resulting in a sequence of RL pairs. The RL pairs having highest frequency are chosen for embedding data. The threshold for black run-length and run-length pairs are set for data embedding. At the receiver side the encrypted message is extracted and it is decompressed in the reverse way to as done during embedding process. Then the data is decrypted by using key to get secret data.

1.3.1 Compressed data preparation

The input text message which is to be hidden is converted into bit stream. This bit-stream is encrypted by key at first level of security. Then it is represented in two dimensional matrixes. Since this matrix is sparse, the location where '1' bit is present in the matrix that locations are only considered and it is transformed from two dimensional to one dimensional matrix. This transformation provides security at second level. This transformed compressed data is further compressed by arithmetic compression algorithm by providing security at third level. This process of compressed data preparation minimizes the bits to be embedded and hence increases the hiding capacity.

1.3.2 Information hiding process

Figure 1 depicts the process of hiding information. Binary images are represented by the numbers 0 and 1 corresponding to the white pixel and black pixel, respectively. Identification of black-and-white RL histograms

and formation of the RL pairs of the binary image as described in the table I,II,III. The information to be hidden is encrypted with shared key and represented in encrypted and compressed as described in earlier section. The threshold for black run length RL and length of RL pairs are defined and used for hiding information. Consider embedding data in black RL equal to 1, i.e., assign T to 1. Black pixels having run length one, i.e. T, “0” information bit is embedded and to embed “1” bit of information black pixels having run length two i.e. T+1 is used. Apart from this, threshold for length of RL pairs is also set to embed and extract correct information.

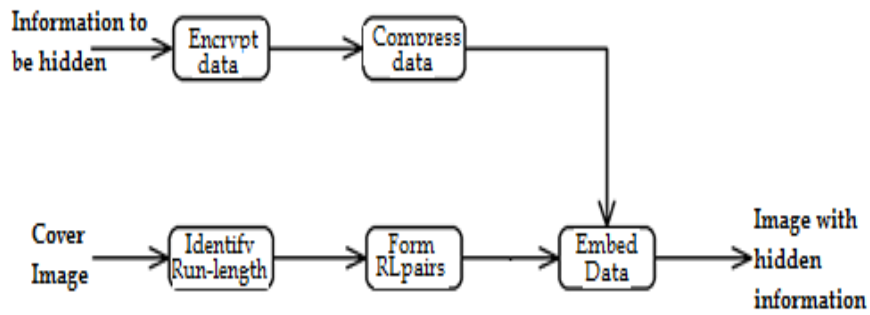


Figure 1. Data embedding process

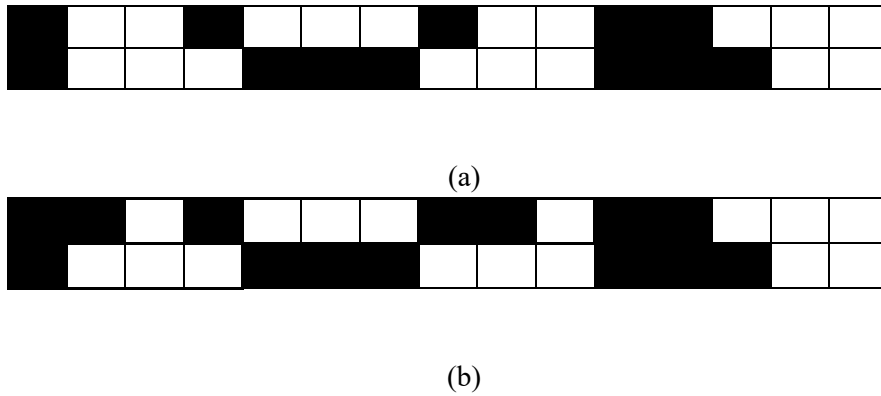


Figure 2. Image part before embedding (a) and after embedding '1010'(b)

TABLE I Image representation as bits ‘1’ and ‘0’ corresponding to binary image and binary image with hidden data

Image	Col →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Original Binary Image	Row 1	1	0	0	1	0	0	0	1	0	0	1	1	0	0	0
	Row 2	1	0	0	0	1	1	1	0	0	0	1	1	1	0	0
Binary Image with hidden data	Row 1	1	1	0	1	0	0	0	1	1	0	1	1	0	0	0
	Row 2	1	0	0	0	1	1	1	0	0	0	1	1	1	0	0

TABLE II Original image run-length pairs and histogram

Original binary image RL pairs	Run-length (RL)	Black RL histogram	White RL histogram
[1-2],[1-3],[1-2], [2,3], [1,3],[3,3], [3,2]	1	4	0
	2	1	3
	3	2	4

TABLE III Binary image with hidden data and corresponding histogram

Binary image after embedding 1010 bits with T=1 and T1=4	Run-length (RL)	Black RL histogram	White RL histogram
[2-1],[1-3],[2-1], [2-3], [1,2],[3,3], [3-2]	1	2	2
	2	3	2
	3	2	3

Assign a parameter T equal to 1, i.e., embed data into black RL equal to 1. In order to embed data, a pair of histograms is created. The sum of the black RL and the white RL in this RL pair remains unchanged. In this example pairs of

length 3 and 4 are considered for hiding information. Since $h(1)=4$, four bits can be embedded and assumed four bits are as follows: 1,0,1,0. Then the pixel at row x1 and column 2 is changed from 0 to 1 in order to change the black RL from 1 to 2. No change is made while embedding 0. Thus 4 bits are embedded by above method.

1.3.3 Information extraction

An alternative black-and-white RL sequence is formed by scanning the binary image from left to right and from top to bottom. An RL pair is formed by combining one black RL and its next white RL, resulting in a series of RL pairs. RL pairs of set threshold are used for information embedding.

The information extraction process is shown in figure 3. The same sequencing as used for information embedding in reverse is used for information extraction. As per the set thresholds information is extracted. As shown in table I, when a black RL of length one is encountered, a bit “0” is extracted, when a black RL of length two is encountered, a bit “1” is extracted. In this way hidden secret information is extracted. This extracted information is decompressed and decrypted to get original information.

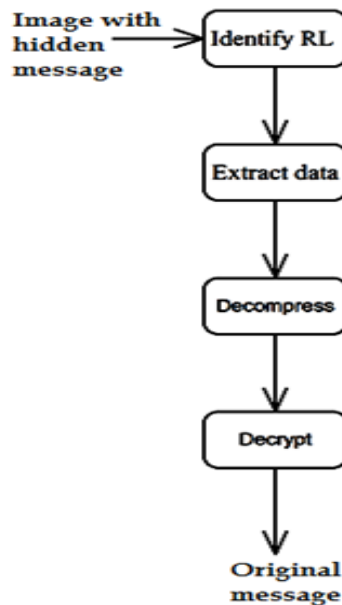


Figure 3. Information extraction process

1.5 Algorithms

1.4.1 Arithmetic coding algorithm

Using this algorithm, maximum compression of the bits is successfully achieved as compared to other compression techniques such as Huffman coding etc. In contrast to Huffman coding, arithmetic coding does not compress each symbol with a discrete number of bits. For every source, it reaches almost the optimum compression in terms of Shannon's theorem and is well suited to adaptive models.

Algorithm:

-
- Step 1. Calculate the probability of each of the symbols.
 - Step 2. Calculate the range_low and range_high of each of these symbols.
 - Step 3. Calculate l and h using the formula:
 $l = \text{low} + \text{range_low}(S);$
 $h = \text{high} + \text{range_high}(S);$
 - Step 4. Generate a compressed code for the data
-

1.4.2 Embedding algorithm

In this algorithm, the concept of run length histogram modification is used in which the maximum number of particular run length are calculated and the value to the threshold is assigned. This threshold then decides in which pixels the information is to be embedded. Two threshold values are assigned in order to prevent maximum distortion to the image and hide the data in the image with maximum efficiency.

Algorithm:

-
- Step 1. Scan the cover image from left to right and top to bottom.
- Step 2. Calculate the run length of black and white pixels and identify those black pixels having maximum run length. The maximum run length as threshold value (T) is considered
- Step 3. For bit '1' :
 If the run length is T and data bit is '1' then
 Change the white pixel to black to make black
 RL T+1
 Else keep it unchanged
- For bit '0':
 If the run length is T+1 and data bit is '0' then
 Change the black pixel to white to make black
 RL T,
 Else keep it unchanged for bit '1'
- Step 1. Save the cover image.
- Step 2. Stop
-

1.4.3 Embedding algorithm

Algorithm:

-
- Step 1. Scan the cover image from left to right and top to bottom.
- Step 2. The 2 dimensional array by using count to find index of run length having value equal to T or T+1 by following formula:
 $\text{Index} = \text{count} + \text{row number} * \text{width of image}$
- Step 3. After calculating index
 If the run length is T then extract data bit as "0"
 If the run length is T+1 extract data bit as "1"
- Step 4. Stop
-

1.6 Result Analysis

The technique is tested with set of standard images and sample figures are shown below, before and after hiding data and results are analyzed in the figures 4,5,6 and 7 respectively. The PSNR and MSE are used to check the performance of technique



Figure 4. Example tested images (a) Before message hiding (b)After message hiding

TABLE II
RESULT TABLE

Sr. No.	Image	Image size	Total Pixels	Flipped pixel count (no. in bits)	Ratio F/T	MSE in 10^{-4}	PSNR
1	Baboon	225X225	40000	29	0.45	4.43	33.54
2	Lena	256X256	88205	37	0.58	5.65	32.48
3	Pillar	256X256	65536	33	0.52	5.04	32.98
4	Cartoon1	257X196	83400	31	0.48	6.15	32.11
5	Image 1	256X256	88205	23	0.45	4.43	33.54
6	Image 2	256X256	88205	30	0.58	5.65	32.48
7	Image 3	256X256	88205	39	0.52	5.04	32.98
8	Image 4	256X256	88205	32	0.48	6.15	32.11

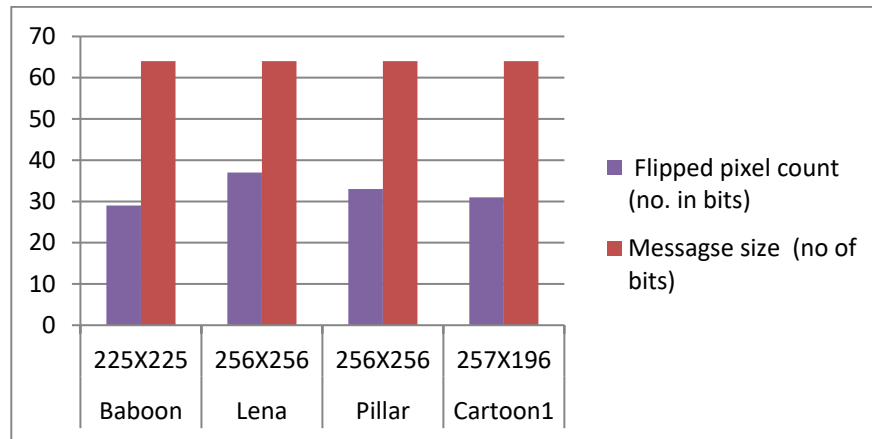


Figure 5. Comparison of Size of message embedded versus flipped pixels

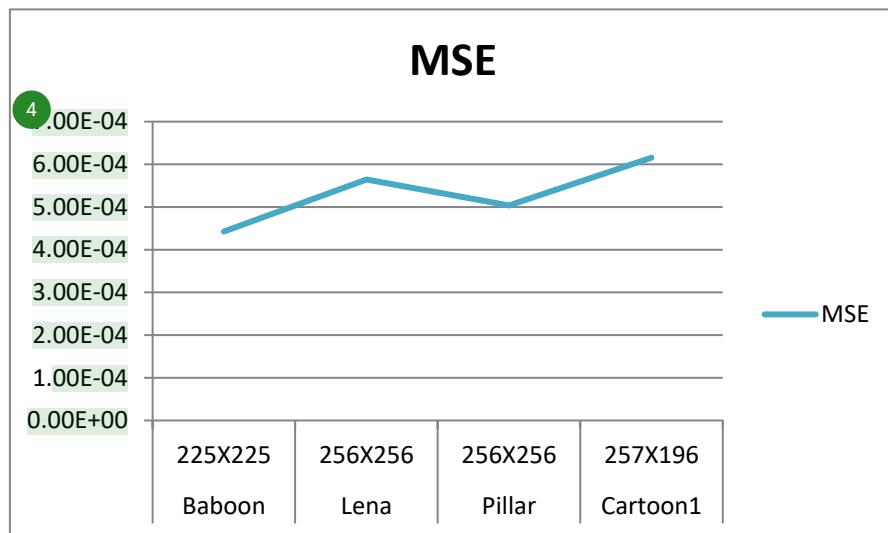


Figure 6. Performance analysis using MSE

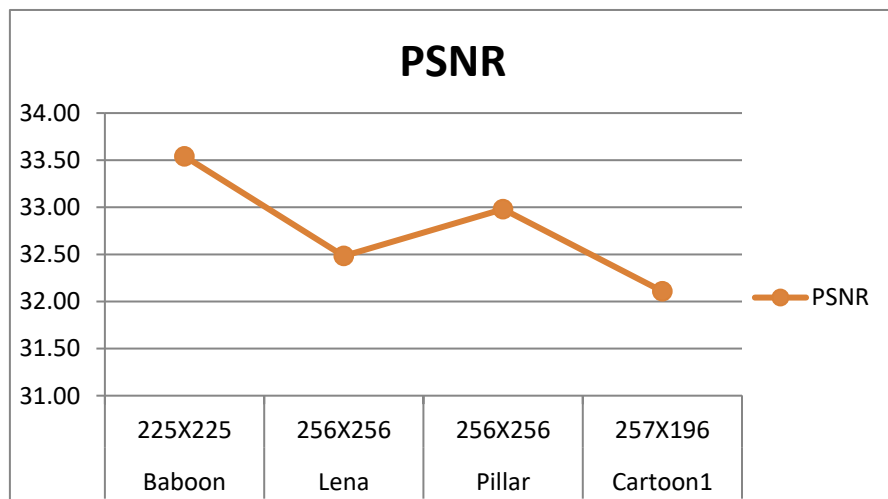


Figure 7 . Performance analysis using PSNR

1.7 Discussion

We have tested the set of standard images using the MSE and PSNR performance matrix. The table 2 shows test results for images in figure 4(a). The images with hidden data are shown in figure 4(b). The table presents the information of image as size, embeddable blocks, flipped pixel count, flipped pixel count to size of hidden data, MSE, PSNR for the hidden data “success” and key “abc”. Total encrypted hidden data has the length of 64 bits. The data is hidden sequentially in blocks satisfying the embeddability test. Result shows that about 50 % of hidden data embedded without flipping pixel. The embeddability will be more better, if the images has more number of RL pairs satisfying threshold set. The performance of any data hiding technique is highly depends on the pattern of cover image and sequence of bits in data to be hidden. The pattern of RL pairs in cover image and the number of RL pairs satisfying embeddability criteria matters for the performance of our technique. The number of flipped pixels is dependent on run length of black and white pixel pairs and message data bit pattern matching. More the match less the visual distortion. The hiding capacity can be enhanced setting other threshold values based on image pattern.

1.8 Conclusion

The various techniques available for data hiding have been analyzed and studied. This paper proposes a novel sparse matrix based secured data encryption scheme for binary image data hiding using RL Histogram modification. Thus using this technique, the hidden data can be encrypted and also can be embedded in the image with maximum efficiency and minimum distortion to the image and thus the hidden data from the image is extracted. In this proposed method, combination of encryption with data embedding is done using run-length histogram modification. This algorithm has reasonable imperceptibility and can be used for secret data transmission, according to experiments. This algorithm has a high correlation with subjective assessment, according to experiments. It can be

6
 applied in a wide range of situations range of applications involving visual distortion in digital binary document images, such as lossless compression and data hiding.

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